

Closing the Rvachev–Lagrange Loop

**Exact fixed-radius polynomial synthesis, signed Markov decoders,
coefficient-space projectors, and dyadic spectral multiplicities**

Research report for Vladimir Reshetnikov’s ProveIt project

29 August 2026

Abstract

Let $u = u_p$ be Rvachev’s compactly supported C^∞ function, normalized as an even probability density on $[-1, 1]$. The repository volume *Exact Polynomial Synthesis by Finite Rvachev Up-Function Dictionaries* proves that the integer-shift system generated by a scale- 2^n copy of u reproduces every polynomial of degree at most n , with coefficients obtained by finite inverse-moment Appell deconvolution. Affine conjugation turns that result into the fixed-physical-radius identity

$$p(x) = 2^{-n} \sum_{k=-2^{n+1}+1}^{2^{n+1}-1} [\mathbf{M}_u(D)^{-1}p](k2^{-n}) u(x - k2^{-n}), \quad -1 \leq x \leq 1,$$

for every $p \in \mathcal{P}_n$. Thus the atoms are literal translates of the *same* u ; no dilation remains. Applied to the Lagrange cardinal polynomials ℓ_j , this gives a completely explicit finite representation, whose coefficients are rational combinations of Bernoulli numbers, complete Bell polynomials, and elementary symmetric polynomials of the shifted nodes.

Substituting these formulas into the Lagrange interpolant of u closes the loop. The resulting up-shift sum is not a new approximation operator in disguise: at every fixed degree it factors the ordinary interpolation projector exactly. In operator notation,

$$\mathcal{I}_n = \mathcal{R}_n C_n \mathcal{E}_n, \quad B_n C_n = I_{n+1}, \quad P_n = C_n B_n,$$

where C_n converts nodal data to up-amplitudes and B_n samples an up-dictionary at the interpolation nodes. After a natural normalization, B_n is a positive row-stochastic matrix and the Appell–Lagrange decoder is a signed row-stochastic right inverse. Consequently P_n is an exact oblique projector with eigenvalue 1 of multiplicity $n + 1$ and eigenvalue 0 of multiplicity $4 \cdot 2^n - n - 2$. Its zero eigenspace splits into an explicit Thue–Morse/2-adic alias sector of dimension $2^{n+1} - n - 2$ and a sample-invisible local-spline sector of dimension 2^{n+1} . Nested node sets lift to a commuting flag of coefficient projectors; for dyadic degrees $n_q = 2^q$, the detail idempotents have eigenvalue-1 multiplicities 2^{q-1} .

The factorization proves an exact transfer principle: every finite truncation error inside a level is zero, and the infinite up-shift series converges, diverges, or exhibits Runge oscillation precisely when the underlying Lagrange series does. Chebyshev–Lobatto levels therefore give superalgebraic convergence for u , while the equispaced loop inherits the repository’s Runge instability. Exact rational computations verify the right-inverse identity through degree 8; reduced-QR experiments show the projector norm growing from 2.19 at degree 2 to 3.02×10^3 at degree 12 for equispaced nodes, versus 1.11×10^2 for Chebyshev nodes. A concrete first closed loop is

$$1 - x^2 = \sum_{k=-7}^7 \left(\frac{5}{18} - \frac{k^2}{64} \right) u\left(x - \frac{k}{4}\right), \quad -1 \leq x \leq 1.$$

The report ends with conjectures on fixed-radius complexity, projector obliquity, minimal-variation alias gauges, and Lambert- W /log-periodic asymptotics of the equispaced boundary layer.

Contents

1	Corpus audit, provenance, and novelty boundary	4
2	Notation and imported structural facts	4
2.1	The up-function as a probability density	4
2.2	The inverse-moment Appell operator	5
2.3	Lagrange interpolation	6
3	Affine conjugation: exact synthesis by literal shifts	6
3.1	A compressed but asymmetric unscaled basis	7
4	Lagrange cardinal polynomials as finite up-shift combinations	8
4.1	A closed coefficient formula in elementary symmetric polynomials	8
4.2	Rationality and symmetry on dyadic combs	9
4.3	The first nontrivial closed identity	9
5	Closing the loop	9
5.1	The fixed-level self-representation	9
5.2	The Lagrange series becomes an up-shift series	10
6	Operator factorization and the signed Markov inverse	11
6.1	Synthesis, sampling, and coefficient conversion	11
6.2	The coefficient-space projector	12
6.3	Trace and determinant sum rules	13
6.4	Why negative decoder weights are forced	13
7	Local shift-space geometry and the spectral multiplicity ledger	14
7.1	Polynomial and nodal-zero summands	14
7.2	Alias kernel and exact sequence	15
7.3	Explicit 2-adic zero modes and Appell one modes	15
7.4	What compression removes	16
8	Nested projector flags and dyadic detail multiplicities	16
9	Convergence, Runge behavior, and coordinate stability	17
9.1	An exact transfer theorem	17
9.2	Chebyshev–Lobatto convergence	17
9.3	Equispaced Runge behavior	18
9.4	Lebesgue growth must appear in the decoder	18
9.5	Computational complexity	18
10	Numerical experiments	19
10.1	Exact rational right-inverse checks	19
10.2	Projector obliquity without forming the projector	19
10.3	The loop inherits interpolation error	19
10.4	Reproducibility files	21
11	Further deductions, conjectures, and research directions	21
11.1	Minimal fixed-radius complexity	22
11.2	Asymptotics of projector obliquity	22
11.3	Alias gauges and minimum-variation exact synthesis	22
11.4	A continuum limit of the signed Markov decoder	23

11.5	q -binomial minors and Thue–Morse spectral algebra	23
11.6	Lambert- W and the Runge boundary layer	23
11.7	Node sets co-designed with the up kernel	24
11.8	Formalization	24
12	Conclusion	24
A	Formula sheet	25
B	Dimension and multiplicity ledger	25
C	Notes on numerical verification	25
D	Selected corpus crosswalk	26

1 Corpus audit, provenance, and novelty boundary

The analysis uses the recursive documentation snapshot of VladimirReshetnikov/ProveIt at commit

4544e9af6b94820402c69967be1dff3f85e43c1e

from 29 August 2026. The current tree contains 65 $\text{\TeX}$ documents under Analysis/FabiusFunction/docs, including the living exposition, archived derivations, seven paper transcriptions, the Lean walkthrough, the twelve-document Fourier-decay corpus, and the large consolidated frontier volumes. The audit treated archived files as mathematical sources while mapping frozen or absorbed copies to their current canonical expositions.

The two directly controlling documents are:

1. *Exact Polynomial Synthesis by Finite Rvachev Up-Function Dictionaries* (the current drafts/representations/Up_Polynomial_Synthesis/ volume), which proves the scale classification, global Appell deconvolution formula, exact local dimensions, dyadic alias annihilator, endpoint basis, and output-sensitive coefficient algorithm;
2. *Dyadic-Comb Frontiers for the Fabius–Rvachev System* (the current drafts/inverse-and-sampling/Dyadic_Comb_Frontiers/ volume), which sets up global Lagrange interpolation of F and u , exact symmetry compressions, endpoint-flat Hermite variants, and the Runge/stability study.

The primary exposition supplies the normalization of F and u , the Fourier sinc product, functional equations, moments, and exact dyadic values. The representation, exponent/ q -series, Thue–Morse, inverse/sampling, spectral/arithmetic, and Fourier-decay frontier volumes were checked for nearby formulas and for possible duplication.

What is imported, and what is new here

The scale- 2^n polynomial-synthesis theorem, its Appell coefficient operator, the annihilator $A_n(z) = \prod_{r=1}^n [2^r]_z$, and the local-dimension theorem are imported from the synthesis volume. The ordinary interpolation projector and its Runge analysis are imported from the dyadic-comb volume. The contribution of this report is their affine fixed-radius specialization and systematic composition: the explicit Lagrange-cardinal transform, the exact loop factorization, the stochastic/right-inverse normalization, the coefficient projector and its spectral ledger, the function-space direct sum and exact sequence, the nested dyadic projector flag, the decoder-sign/variation bounds, and the associated convergence and stability transfer principles. Searches of the pinned corpus for “coefficient-space projector” and “self-representation operator” found no pre-existing treatment. Novelty claims below are therefore relative to this audited repository snapshot, not unqualified claims of world priority.

The classical fact that atomic-function spaces contain polynomials goes back to the Rvachevs [3, 5]. The present formulas also sit naturally within Strang–Fix theory and the theory of principal shift-invariant spaces [8, 9, 10].

2 Notation and imported structural facts

2.1 The up-function as a probability density

Throughout, $u = \text{up}$ is even, nonnegative, supported on $[-1, 1]$, and normalized by

$$\int_{-1}^1 u(x) \, dx = 1, \quad u(0) = 1, \quad u(\pm 1) = 0. \quad (2.1)$$

With the Fourier convention $\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} dx$,

$$\widehat{u}(\xi) = \prod_{j \geq 0} \operatorname{sinc}_{\pi} \left(\frac{\xi}{2^j} \right), \quad \operatorname{ord}_{\xi=q} \widehat{u} = 1 + \nu_2(q) \quad (q \in \mathbb{Z} \setminus \{0\}). \quad (2.2)$$

Equivalently, if $Z = \sum_{j \geq 1} 2^{-j} V_j$ with independent $V_j \sim \operatorname{Unif}[-1, 1]$, then u is the density of Z . Its moment generating function is

$$M_u(z) = \mathbb{E} e^{zZ} = \prod_{j \geq 1} \frac{\sinh(z/2^j)}{z/2^j}. \quad (2.3)$$

The logarithm has even cumulants

$$\log M_u(z) = \sum_{r \geq 1} \kappa_{2r} \frac{z^{2r}}{(2r)!}, \quad \kappa_{2r} = \frac{B_{2r}}{2r(1-4^{-r})}, \quad \kappa_{2r+1} = 0. \quad (2.4)$$

2.2 The inverse-moment Appell operator

Define rational numbers γ_r by

$$\frac{1}{M_u(z)} = \sum_{r \geq 0} \gamma_r \frac{z^r}{r!} = \exp \left(- \sum_{s \geq 1} \kappa_{2s} \frac{z^{2s}}{(2s)!} \right). \quad (2.5)$$

Thus $\gamma_{2r+1} = 0$ and

$$\gamma_0 = 1, \quad \gamma_2 = -\frac{1}{9}, \quad \gamma_4 = \frac{31}{675}, \quad \gamma_6 = -\frac{2347}{59535}, \quad \gamma_8 = \frac{1904369}{32531625}. \quad (2.6)$$

The complete Bell-polynomial formula is

$$\gamma_r = Y_r(-\kappa_1, -\kappa_2, \dots, -\kappa_r), \quad (2.7)$$

or, more explicitly,

$$\frac{\gamma_{2r}}{(2r)!} = \sum_{\substack{k_1+2k_2+\dots+rk_r=r \\ k_s \geq 0}} \prod_{s=1}^r \frac{1}{k_s!} \left(-\frac{\kappa_{2s}}{(2s)!} \right)^{k_s}. \quad (2.8)$$

For a polynomial p define the terminating differential operator

$$\mathcal{A}p := M_u(D)^{-1}p = \sum_{r=0}^{\lfloor \deg p/2 \rfloor} \frac{\gamma_{2r}}{(2r)!} p^{(2r)}. \quad (2.9)$$

The deconvolution Appell polynomials are

$$\mathcal{A}_r(y) := \mathcal{A}(y^r) = \sum_{s=0}^r \binom{r}{s} \gamma_s y^{r-s}, \quad \mathcal{A}'_r(y) = r \mathcal{A}_{r-1}(y). \quad (2.10)$$

The operator is triangular with unit diagonal on every polynomial space; hence it is an automorphism of $\mathcal{P}_n$.

2.3 Lagrange interpolation

Let $X_n = \{x_0, \dots, x_n\} \subset [-1, 1]$ consist of distinct nodes and put

$$\omega_n(x) = \prod_{j=0}^n (x - x_j), \quad \ell_j(x) = \frac{\omega_n(x)}{(x - x_j)\omega'_n(x_j)}. \quad (2.11)$$

Then $\ell_j(x_i) = \delta_{ij}$ and

$$(\mathcal{I}_n f)(x) = \sum_{j=0}^n f(x_j) \ell_j(x) \quad (2.12)$$

projects functions onto $\mathcal{P}_n$. We distinguish the evaluation map $\mathcal{E}_n f = (f(x_0), \dots, f(x_n))^T$ from the reconstruction map $\mathcal{L}_n y = \sum_j y_j \ell_j$, so

$$\mathcal{I}_n = \mathcal{L}_n \mathcal{E}_n, \quad \mathcal{E}_n \mathcal{L}_n = I_{n+1}. \quad (2.13)$$

3 Affine conjugation: exact synthesis by literal shifts

The synthesis volume works with the normalized atoms

$$\psi_{\rho,k}(t) = \frac{1}{\rho} u\left(\frac{t-k}{\rho}\right). \quad (3.1)$$

For integer ρ , their Strang–Fix reproduction degree is exactly $\nu_2(\rho)$, because the least multiplicity of a nonzero dual-lattice zero of $\hat{u}(\rho\xi)$ is $1 + \nu_2(\rho)$. The smallest common scale that reproduces $\mathcal{P}_n$ is therefore $m = 2^n$.

Theorem 3.1 (Canonical fixed-radius polynomial synthesis). *Let $p \in \mathcal{P}_n$, put*

$$m = 2^n, \quad h = m^{-1} = 2^{-n}, \quad K_n = \{-2m + 1, -2m + 2, \dots, 2m - 1\}. \quad (3.2)$$

Then the global locally finite identity

$$p(x) = h \sum_{k \in \mathbb{Z}} (\mathcal{A}p)(kh) u(x - kh) \quad (3.3)$$

holds for every $x \in \mathbb{R}$. On $[-1, 1]$ it reduces exactly to the finite sum

$$p(x) = h \sum_{k \in K_n} (\mathcal{A}p)(kh) u(x - kh), \quad -1 \leq x \leq 1. \quad (3.4)$$

Thus $4m - 1 = 4 \cdot 2^n - 1$ literal translates of the same unscaled u form a canonical finite dictionary for every degree- n polynomial on $[-1, 1]$.

Proof. The imported global synthesis theorem says that for $P \in \mathcal{P}_n$,

$$P(t) = \frac{1}{m} \sum_{k \in \mathbb{Z}} Q(k) u\left(\frac{t-k}{m}\right), \quad Q = M_u(mD_t)^{-1} P. \quad (3.5)$$

Set $P(t) = p(t/m)$ and then $t = mx$. Since $D_t^r P(t) = m^{-r} p^{(r)}(t/m)$,

$$Q(t) = \sum_r \gamma_r \frac{m^r D_t^r P(t)}{r!} = (\mathcal{A}p)(t/m).$$

Substitution in (3.5) gives (3.3). If $x \in [-1, 1]$, the factor $u(x - kh)$ can be nonzero only when $kh \in [x - 1, x + 1] \subset [-2, 2]$. The boundary indices $k = \pm 2m$ contribute $u(\pm 1) = 0$, leaving exactly K_n . $\square$

Remark 3.2 (Why this is genuinely a no-dilation formula). The exponential scale $m = 2^n$ has not disappeared; it has moved from the *shape* to the *spacing*. In the original variable, the atom is $u((t-k)/m)$. Under $t = mx$, it becomes $u(x - k/m)$: the support radius is literally one, while the grid spacing is $h = 2^{-n}$. This is precisely the physical-radius- $\rho = 1$ regime requested in the problem.

Corollary 3.3 (Monomial reproduction). *For $0 \leq r \leq n$,*

$$x^r = h \sum_{k \in K_n} \mathcal{A}_r(kh) u(x - kh), \quad -1 \leq x \leq 1. \quad (3.6)$$

In particular,

$$1 = h \sum_{k \in K_n} u(x - kh), \quad (3.7)$$

$$x = h \sum_{k \in K_n} (kh) u(x - kh), \quad (3.8)$$

$$x^2 = h \sum_{k \in K_n} \left((kh)^2 - \frac{1}{9} \right) u(x - kh). \quad (3.9)$$

The first identity says that the oversampled translates form a partition of unity after multiplication by h ; it will turn the collocation matrix into a Markov matrix in [section 6](#).

3.1 A compressed but asymmetric unscaled basis

The canonical formula (3.4) is redundant. The multi-cell dimension theorem in the synthesis volume yields a smaller unscaled basis after the same physical specialization.

Theorem 3.4 (Alias-compressed fixed-radius basis). *With $m = 2^n$ and $h = 1/m$, define*

$$\tau_r = -(n+1)h + rh, \quad 0 \leq r \leq 2m + n. \quad (3.10)$$

The $2m + n + 1$ functions $u(x - \tau_r)$ restrict to a basis of the complete local shift space generated on $[-1, 1]$ by all active $h\mathbb{Z}$ -translates. Every $p \in \mathcal{P}_n$ therefore has a unique representation in this basis. A reflected left-end basis is obtained by $x \mapsto -x$.

Proof. Apply the imported physical multi-cell theorem with interval $[a, b] = [-1, 1]$, polynomial degree n , scale parameter $M = m$, and $J = 2m$ cells. Then $h = (b - a)/J = 1/m$ and the physical atom radius is $Mh = 1$. The theorem's basis centers are

$$a + h(M - n - 1 + r) = -1 + \frac{m - n - 1 + r}{m} = -(n+1)h + rh.$$

After which its dimension formula gives $J + n + 1 = 2m + n + 1$. □

The coefficient reduction from $4m - 1$ to $2m + n + 1$ is governed by the dyadic alias annihilator

$$A_n(z) = \prod_{r=1}^n [2^r]_z = \prod_{r=1}^n \frac{1 - z^{2^r}}{1 - z} = \prod_{\ell=1}^n (z^{2^{\ell-1}} + 1)^{n-\ell+1}. \quad (3.11)$$

Its degree is $2m - n - 2$, exactly the number of dense-dictionary aliases eliminated. This is where the q -integer and Thue–Morse structure enters the fixed-radius problem.

Proposition 3.5 (Optimal uniform-lattice spacing). *Among uniform lattices $h\mathbb{Z}$ for which $1/h$ is an integer, global reproduction of $\mathcal{P}_n$ by unscaled translates $u(x - kh)$ requires $h \leq 2^{-n}$. Hence the grid in [theorem 3.1](#) is the coarsest uniform integer-ratio grid with degree- n Strang–Fix order.*

Proof. Rescale by $t = x/h$. The generator becomes $u((t - k)/\rho)$ with $\rho = 1/h$. The imported scale classification gives reproduction degree $\nu_2(\rho)$, so $\nu_2(1/h) \geq n$ and therefore $1/h$ is divisible by 2^n . □

4 Lagrange cardinal polynomials as finite up-shift combinations

The fixed-radius theorem may now be applied one cardinal at a time. This is the requested representation on $[-1, 1]$.

Theorem 4.1 (Canonical up-shift representation of the Lagrange basis). *For each $0 \leq j \leq n$ and every $x \in [-1, 1]$,*

$$\ell_j(x) = h \sum_{k \in K_n} D_{kj}^{(n)} u(x - kh), \quad D_{kj}^{(n)} := (\mathcal{A} \ell_j)(kh). \quad (4.1)$$

Equivalently,

$$D_{kj}^{(n)} = \sum_{r=0}^{\lfloor n/2 \rfloor} \frac{\gamma_{2r}}{(2r)!} \ell_j^{(2r)}(kh). \quad (4.2)$$

All sums are finite. If the nodes are rational, every coefficient is rational.

Proof. Apply [theorem 3.1](#) to $p = \ell_j$. Formula (4.2) is the terminating definition of $\mathcal{A}$. Rationality follows from the rational Bernoulli numbers, hence rational γ_{2r} , and from the rational coefficients of a Lagrange cardinal at rational nodes. $\square$

4.1 A closed coefficient formula in elementary symmetric polynomials

The derivative form is already constructive, but it admits a compact node-symmetric expression. For a finite multiset Y , let $e_r(Y)$ denote the elementary symmetric polynomial of degree r , with $e_0 = 1$ and $e_r = 0$ outside the natural range.

Proposition 4.2 (Bernoulli–Appell–symmetric coefficient formula). *Let $t_k = kh$. Then*

$$D_{kj}^{(n)} = \frac{1}{\omega'_n(x_j)} \sum_{r=0}^{\lfloor n/2 \rfloor} \gamma_{2r} e_{n-2r}(\{t_k - x_s : s \neq j\}). \quad (4.3)$$

Consequently every coefficient is a finite Bell–Bernoulli polynomial in the cumulants (2.4), followed by one elementary-symmetric evaluation on the dyadic grid.

Proof. Introduce an auxiliary variable z . From (2.11),

$$\ell_j(t_k + z) = \frac{1}{\omega'_n(x_j)} \prod_{s \neq j} (t_k - x_s + z).$$

The coefficient of z^q is $e_{n-q}(\{t_k - x_s : s \neq j\})/\omega'_n(x_j)$, and it also equals $\ell_j^{(q)}(t_k)/q!$. Substitute $q = 2r$ into (4.2). $\square$

For equispaced nodes $x_j = -1 + 2j/n$,

$$\omega'_n(x_j) = \left(\frac{2}{n}\right)^n (-1)^{n-j} j!(n-j)!, \quad (4.4)$$

and, after setting $s = n(x + 1)/2$,

$$\ell_j(x) = \binom{s}{j} \binom{n-s}{n-j}. \quad (4.5)$$

Thus the usual $q = 1$ Pascal row is composed with the q -integer dyadic annihilator (3.11) and the Bernoulli–Appell deconvolution (2.9). For affine geometric nodes $x_j = a + bq^j$, the normalizing derivative becomes

$$\omega'_n(x_j) = b^n (-1)^j q^{jn-j(j+1)/2} (q; q)_j (q; q)_{n-j}, \quad (4.6)$$

so the same transform acquires the q -Pochhammer denominators familiar from the geometric Lagrange formulas elsewhere in the corpus.

4.2 Rationality and symmetry on dyadic combs

Corollary 4.3 (Exact rational loops). *Suppose all nodes are dyadic rationals. Then every $D_{kj}^{(n)}$ is rational. If the data $f(x_j)$ are rational—in particular, if $f = u$ and the nodes are dyadic—then every amplitude in the closed loop of [section 5](#) is rational.*

The statement uses the established rationality of the Fabius/up values at dyadic arguments. It is especially useful for the equispaced degrees $n = 2^q$, where $x_j = -1 + 2j/n$ is a dyadic comb while the reproduction grid is the much finer grid $2^{-n}\mathbb{Z}$.

Proposition 4.4 (Symmetric coefficient compression). *Assume the node set is symmetric, $x_{n-j} = -x_j$, and the interpolated data are even. Then $p = \mathcal{I}_n f$ is even, $\mathcal{A}p$ is even, and the canonical amplitudes satisfy $c_{-k} = c_k$. Therefore*

$$p(x) = c_0 u(x) + \sum_{k=1}^{2m-1} c_k (u(x - kh) + u(x + kh)). \quad (4.7)$$

If n is odd, the even interpolant has degree at most $n - 1$, so the entire sum may instead be synthesized with the coarser grid $2^{-(n-1)}\mathbb{Z}$, halving the canonical atom count.

Proof. Uniqueness of interpolation gives $p(-x) = p(x)$. The operator $\mathcal{A}$ uses only even derivatives and preserves evenness, proving $c_{-k} = c_k$. An even polynomial cannot have odd degree, which gives the degree drop when n is odd. $\square$

4.3 The first nontrivial closed identity

Example 4.5 (Degree two). Take the nodes $-1, 0, 1$. Since $u(\pm 1) = 0$ and $u(0) = 1$,

$$(\mathcal{I}_2 u)(x) = 1 - x^2. \quad (4.8)$$

Here $m = 4$, $h = 1/4$, and

$$\mathcal{A}(1 - x^2) = 1 - x^2 - \frac{1}{18}(-2) = \frac{10}{9} - x^2.$$

Consequently

$$1 - x^2 = \sum_{k=-7}^7 \left(\frac{5}{18} - \frac{k^2}{64} \right) u\left(x - \frac{k}{4}\right), \quad -1 \leq x \leq 1. \quad (4.9)$$

The negative coefficients near $k = \pm 7$ are not numerical artifacts; signed amplitudes are structurally unavoidable once overlapping positive kernel rows must be inverted, as shown in [theorem 6.7](#).

For comparison, the degree-four equispaced interpolant through $u(-1), u(-1/2), u(0), u(1/2), u(1)$ is

$$p_4(x) = 1 - \frac{7}{3}x^2 + \frac{4}{3}x^4, \quad \mathcal{A}p_4(x) = \frac{2674}{2025} - \frac{29}{9}x^2 + \frac{4}{3}x^4, \quad (4.10)$$

so p_4 has an exact 63-shift rational representation at spacing $1/16$.

5 Closing the loop

5.1 The fixed-level self-representation

Let $y_j = f(x_j)$. Substitute (4.1) into the Lagrange formula (2.12):

$$\begin{aligned} (\mathcal{I}_n f)(x) &= \sum_{j=0}^n y_j \ell_j(x) \\ &= h \sum_{k \in K_n} \left(\sum_{j=0}^n D_{kj}^{(n)} y_j \right) u(x - kh). \end{aligned} \quad (5.1)$$

This gives the central loop theorem.

Theorem 5.1 (Exact finite Rvachev–Lagrange loop). *For arbitrary data $y = (y_0, \dots, y_n)^T$, define*

$$c_k(y) = h \sum_{j=0}^n D_{kj}^{(n)} y_j = h [\mathcal{A}(\mathcal{L}_n y)](kh). \quad (5.2)$$

Then

$$(\mathcal{L}_n y)(x) = \sum_{k \in K_n} c_k(y) u(x - kh), \quad -1 \leq x \leq 1. \quad (5.3)$$

In particular, for $y_j = u(x_j)$,

$$(\mathcal{I}_n u)(x) = \sum_{k \in K_n} c_{n,k} u(x - kh), \quad c_{n,k} = h [\mathcal{A}(\mathcal{I}_n u)](kh). \quad (5.4)$$

The right-hand side is an exact self-representation of the interpolant of u . No assertion that u itself is a finite sum is being made.

Proof. Equation (5.3) is (5.1); the second expression in (5.2) follows from linearity of $\mathcal{A}$. Set $y_j = u(x_j)$ for (5.4). $\square$

Remark 5.2 (Cardinal substitution versus post-summation synthesis). Synthesizing each degree- n cardinal and then summing is algebraically identical to first forming $p = \mathcal{I}_n f$ and synthesizing p . The latter can be cheaper when symmetry forces a degree drop, because the reproduction parameter may be reduced to the actual degree of p . This distinction is important in exact arithmetic: the termwise formula is canonical for the transform matrix, while post-summation synthesis is canonical for the final polynomial.

5.2 The Lagrange series becomes an up-shift series

Let

$$X^{(0)} \subset X^{(1)} \subset \dots, \quad |X^{(q)}| = n_q + 1, \quad n_0 < n_1 < \dots, \quad (5.5)$$

be nested node sets, and let $\mathcal{I}^{(q)}$ denote the corresponding interpolation projector. Whenever the Lagrange interpolants converge,

$$u = \mathcal{I}^{(0)} u + \sum_{q \geq 1} (\mathcal{I}^{(q)} - \mathcal{I}^{(q-1)}) u \quad (5.6)$$

with convergence in the same function norm. Put

$$\Delta_q u := \begin{cases} \mathcal{I}^{(0)} u, & q = 0, \\ (\mathcal{I}^{(q)} - \mathcal{I}^{(q-1)}) u, & q \geq 1. \end{cases} \quad (5.7)$$

Since $\Delta_q u \in \mathcal{P}_{n_q}$, [theorem 3.1](#) gives

$$\Delta_q u(x) = h_q \sum_{k \in K_{n_q}} (\mathcal{A} \Delta_q u)(kh_q) u(x - kh_q), \quad h_q = 2^{-n_q}. \quad (5.8)$$

Substitution in (5.6) produces the closed infinite loop

$$u(x) = \sum_{q \geq 0} h_q \sum_{k \in K_{n_q}} (\mathcal{A} \Delta_q u)(kh_q) u(x - kh_q), \quad (5.9)$$

provided the original interpolation sequence converges. Its partial sum through level Q is *exactly* $\mathcal{I}^{(Q)} u$; therefore

$$\|u - \text{partial sum through } Q\| = \|\mathcal{I}^{(Q)} u - u\| \quad (5.10)$$

for every norm in which both sides are defined.

Warning 5.3 (Double-exponential bookkeeping under degree doubling). If $n_q = 2^q$, then the physical grid spacing in the q th polynomial detail is $h_q = 2^{-2^q}$ and the dense atom count is $4 \cdot 2^{2^q} - 1$. Thus (5.9) is an exact structural identity, not by itself an efficient numerical representation. The compressed basis reduces the leading factor from four to two, but the fixed-radius cost remains exponential in the polynomial degree and therefore double-exponential in a dyadic level index.

6 Operator factorization and the signed Markov inverse

The loop becomes substantially more informative when written as a sequence of finite-dimensional maps.

6.1 Synthesis, sampling, and coefficient conversion

Let $N_n = |K_n| = 4m - 1$. Define

$$(\mathcal{R}_n c)(x) := \sum_{k \in K_n} c_k u(x - kh), \quad \mathcal{R}_n : \mathbb{R}^{N_n} \longrightarrow C[-1, 1], \quad (6.1)$$

$$(\mathcal{S}_n p)_k := h(\mathcal{A}p)(kh), \quad \mathcal{S}_n : \mathcal{P}_n \longrightarrow \mathbb{R}^{N_n}. \quad (6.2)$$

Then [theorem 3.1](#) is simply

$$\mathcal{R}_n \mathcal{S}_n = I_{\mathcal{P}_n}. \quad (6.3)$$

The nodal-to-coefficient matrix is

$$C_n := \mathcal{S}_n \mathcal{L}_n, \quad (C_n)_{kj} = h D_{kj}^{(n)}, \quad (6.4)$$

and the raw collocation matrix is

$$B_n := \mathcal{E}_n \mathcal{R}_n, \quad (B_n)_{ik} = u(x_i - kh). \quad (6.5)$$

The interpolation operator factors as

$$\boxed{\mathcal{I}_n = \mathcal{R}_n C_n \mathcal{E}_n}. \quad (6.6)$$

Theorem 6.1 (Exact right inverse and discrete biorthogonality). *The collocation and coefficient matrices satisfy*

$$\boxed{B_n C_n = I_{n+1}}. \quad (6.7)$$

Equivalently,

$$h \sum_{k \in K_n} u(x_i - kh) (\mathcal{A} \ell_j)(kh) = \delta_{ij}. \quad (6.8)$$

Proof. Using (6.3) and (2.13),

$$B_n C_n = \mathcal{E}_n \mathcal{R}_n \mathcal{S}_n \mathcal{L}_n = \mathcal{E}_n \mathcal{L}_n = I_{n+1}.$$

The component form is (6.8). □

Normalize the matrices by

$$\tilde{B}_n := h B_n, \quad D_n := h^{-1} C_n, \quad (6.9)$$

so that D_n has the entries in (4.2). Then

$$\tilde{B}_n D_n = I_{n+1}. \quad (6.10)$$

Proposition 6.2 (A positive Markov encoder and a signed Markov decoder). *The matrix $\tilde{B}_n$ is nonnegative and row-stochastic, while every row of D_n sums to one:*

$$\tilde{B}_n \mathbf{1}_{N_n} = \mathbf{1}_{n+1}, \quad D_n \mathbf{1}_{n+1} = \mathbf{1}_{N_n}. \quad (6.11)$$

Thus D_n is a signed row-stochastic right inverse of the Markov kernel $\tilde{B}_n$.

Proof. Nonnegativity follows from $u \geq 0$. The row sum of $\tilde{B}_n$ is $h \sum_k u(x_i - kh) = 1$ by (3.7). Since $\sum_j \ell_j = 1$ and $\mathcal{A}1 = 1$, $\sum_j D_{kj} = \mathcal{A}(\sum_j \ell_j)(kh) = 1$. $\square$

The terminology is probabilistic but exact: for each node x_i , $\tilde{B}_{i\cdot}$ is a lattice probability distribution obtained by sampling a translate of the up-density. The signed decoder reverses this smoothing on the finite polynomial data space.

6.2 The coefficient-space projector

Theorem 6.3 (Self-representation projector). *Define*

$$\boxed{P_n := C_n B_n = D_n \tilde{B}_n \in \mathbb{R}^{N_n \times N_n}.} \quad (6.12)$$

Then:

1. $P_n^2 = P_n$;
2. $\text{Ran } P_n = \text{Ran } C_n = \text{Ran } \mathcal{S}_n$ and $\text{Ker } P_n = \text{Ker } B_n$;
3. $\text{rank } P_n = n + 1$;
4. P_n is diagonalizable and

$$\chi_{P_n}(\lambda) = \lambda^{4 \cdot 2^n - n - 2} (\lambda - 1)^{n+1}; \quad (6.13)$$

5. $\text{tr } P_n = n + 1$ and $P_n \mathbf{1} = \mathbf{1}$;
6. the intertwining identity

$$\mathcal{R}_n P_n = \mathcal{I}_n \mathcal{R}_n \quad (6.14)$$

holds on all coefficient vectors.

Proof. By (6.7), $P_n^2 = C_n (B_n C_n) B_n = C_n B_n$. The range and kernel statements are standard consequences of a right-inverse factorization: C_n is injective, B_n is surjective, and $C_n B_n c = 0$ iff $B_n c = 0$. Hence the rank is $n + 1$. An idempotent has minimal polynomial dividing $z(z - 1)$ and is therefore diagonalizable, with the asserted multiplicities. The trace equals the rank. Equation (6.11) gives $P_n \mathbf{1} = D_n \tilde{B}_n \mathbf{1} = D_n \mathbf{1} = \mathbf{1}$. Finally,

$$\mathcal{R}_n P_n = \mathcal{R}_n C_n B_n = \mathcal{L}_n \mathcal{E}_n \mathcal{R}_n = \mathcal{I}_n \mathcal{R}_n.$$

$\square$

The spectrum is deceptively benign: it contains only 0 and 1. Numerical sensitivity is encoded not in eigenvalue moduli but in the angle between $\text{Ran } C_n$ and $\text{Ker } B_n$. In the Euclidean norm, an oblique projection satisfies

$$\|P_n\|_2 = \frac{1}{\sin \theta_n}, \quad (6.15)$$

where θ_n is the smallest principal angle between its range and kernel. The norm can therefore be enormous even though the spectral radius is one.

Corollary 6.4 (Canonicalization of arbitrary amplitudes). *For any coefficient vector c , $P_n c$ is the canonical Bernoulli–Appell coefficient vector of the polynomial interpolating the nodal values of $\mathcal{R}_n c$. It is the unique vector in $\text{Ran } C_n$ with the same nodal samples as c :*

$$B_n P_n c = B_n c, \quad P_n C_n = C_n. \quad (6.16)$$

6.3 Trace and determinant sum rules

The right-inverse identity produces finite identities involving up-values and Appell-deconvolved cardinals.

Corollary 6.5 (Trace identity).

$$h \sum_{k \in K_n} \sum_{j=0}^n u(x_j - kh)(\mathcal{A}\ell_j)(kh) = n + 1. \quad (6.17)$$

Proof. Take the trace of $D_n \tilde{B}_n$. □

Corollary 6.6 (Cauchy–Binet up/Appell minor identity). *For every $(n + 1)$ -subset $S \subset K_n$, let $\tilde{B}_{n, \cdot, S}$ be the corresponding square column minor and $D_{n, S, \cdot}$ the corresponding row minor. Then*

$$1 = \sum_{\substack{S \subset K_n \\ |S|=n+1}} \det(\tilde{B}_{n, \cdot, S}) \det(D_{n, S, \cdot}). \quad (6.18)$$

For dyadic nodes every term is rational.

Proof. Apply the Cauchy–Binet formula to $\det(\tilde{B}_n D_n) = 1$. □

Identity (6.18) is a finite determinantal partition of one, mixing the positive up-kernel minors with signed Bernoulli–Appell/Lagrange minors. Determining its cancellation pattern is an open total-positivity problem discussed in [section 11](#).

6.4 Why negative decoder weights are forced

Theorem 6.7 (Overlap forces signed decoding). *Let B be a nonnegative row-stochastic matrix and let D satisfy*

$$BD = I, \quad D\mathbf{1} = \mathbf{1}. \quad (6.19)$$

If two distinct rows $B_{i, \cdot}$ and $B_{j, \cdot}$ have a common strictly positive column, then D cannot be entrywise nonnegative.

Proof. If $D \geq 0$ and $D\mathbf{1} = \mathbf{1}$, every row $D_{k, \cdot}$ is a probability vector. But

$$e_i = B_{i, \cdot} D = \sum_k B_{ik} D_{k, \cdot}$$

is a convex combination of these rows. Since e_i is an extreme point of the probability simplex, every k with $B_{ik} > 0$ must satisfy $D_{k, \cdot} = e_i$. A common positive column for rows i and j would force the same decoder row to equal both e_i and e_j . □

For the standard equispaced and Chebyshev–Lobatto node sets with $n \geq 2$, neighboring up-kernel rows overlap on many grid columns, so the theorem applies to $B = \tilde{B}_n$. This gives a finite-dimensional convex-geometric explanation for the signed amplitudes seen in (4.9).

Theorem 6.8 (Quantitative total-variation lower bound). *Under (6.19), define*

$$V(D) := \max_k \sum_j |D_{kj}|. \quad (6.20)$$

Then for every pair of distinct rows i, j ,

$$V(D) \geq \frac{2}{\|B_{i, \cdot} - B_{j, \cdot}\|_1} = \frac{1}{\text{TV}(B_{i, \cdot}, B_{j, \cdot})}. \quad (6.21)$$

Moreover some row of D has negative mass at least

$$\frac{1}{\|B_{i,\cdot} - B_{j,\cdot}\|_1} - \frac{1}{2}. \quad (6.22)$$

Proof. Since $(B_{i,\cdot} - B_{j,\cdot})D = e_i - e_j$,

$$2 = \|e_i - e_j\|_1 \leq \sum_k |B_{ik} - B_{jk}| \|D_{k,\cdot}\|_1 \leq \|B_{i,\cdot} - B_{j,\cdot}\|_1 V(D).$$

If a row sums to one, its total negative mass is $(\|D_{k,\cdot}\|_1 - 1)/2$, giving (6.22). $\square$

For equispaced nodes separated by $\delta_n = 2/n$, the normalized rows are Riemann samples of two translates of the up-density. Since u is even and strictly unimodal, their continuum total variation is

$$\text{TV}(u(\cdot), u(\cdot - \delta_n)) = \int_{-\delta_n/2}^{\delta_n/2} u(t) dt = \delta_n + o(\delta_n), \quad (6.23)$$

using $u(0) = 1$. Because $h = 2^{-n} = o(\delta_n)$, the discrete total variation has the same asymptotic, and (6.21) yields the universal bound

$$V(D_n) \geq \frac{n}{2}(1 + o(1)). \quad (6.24)$$

The canonical Appell decoder grows much faster in the experiments, because it also performs polynomial extrapolation from $[-1, 1]$ to grid points near ± 2 .

7 Local shift-space geometry and the spectral multiplicity ledger

Let

$$\mathcal{V}_n := \text{span}\{u(\cdot - kh)|_{[-1,1]} : k \in K_n\}. \quad (7.1)$$

The scaled multi-cell theorem quoted in [theorem 3.4](#) gives

$$\dim \mathcal{V}_n = 2m + n + 1, \quad \mathcal{V}_n \cap \mathbb{R}[x] = \mathcal{P}_n. \quad (7.2)$$

7.1 Polynomial and nodal-zero summands

Define the nodal-zero space

$$\mathcal{Z}_n := \{v \in \mathcal{V}_n : v(x_j) = 0, 0 \leq j \leq n\}. \quad (7.3)$$

Theorem 7.1 (Exact local direct sum).

$$\boxed{\mathcal{V}_n = \mathcal{P}_n \oplus \mathcal{Z}_n, \quad \dim \mathcal{Z}_n = 2m.} \quad (7.4)$$

The restriction of $\mathcal{I}_n$ to $\mathcal{V}_n$ is the projection onto $\mathcal{P}_n$ along $\mathcal{Z}_n$.

Proof. Evaluation on $\mathcal{V}_n$ is surjective because $\mathcal{P}_n \subset \mathcal{V}_n$ and evaluation is an isomorphism from $\mathcal{P}_n$ to $\mathbb{R}^{n+1}$. Hence

$$\dim \mathcal{Z}_n = \dim \mathcal{V}_n - (n + 1) = 2m.$$

The intersection $\mathcal{P}_n \cap \mathcal{Z}_n$ is zero, since a degree- n polynomial with $n + 1$ distinct zeros vanishes identically. The dimensions then give the direct sum. Interpolation fixes $\mathcal{P}_n$ and kills nodal values of $\mathcal{Z}_n$. $\square$

Thus the loop is not merely a representation of one function: it identifies the ordinary interpolation projector as the polynomial/nonpolynomial splitting of a concrete local atomic-function space.

7.2 Alias kernel and exact sequence

The dense synthesis map has

$$\dim \operatorname{Ker} \mathcal{R}_n = N_n - \dim \mathcal{V}_n = 2m - n - 2. \quad (7.5)$$

On the other hand,

$$\dim \operatorname{Ker} B_n = N_n - (n + 1) = 4m - n - 2. \quad (7.6)$$

Theorem 7.2 (Alias/sample exact sequence). *Synthesis induces the short exact sequence*

$$0 \longrightarrow \operatorname{Ker} \mathcal{R}_n \longrightarrow \operatorname{Ker} B_n \xrightarrow{\mathcal{R}_n} \mathcal{Z}_n \longrightarrow 0. \quad (7.7)$$

Consequently the zero eigenspace of $\mathcal{P}_n$ has the exact dimension split

$$\boxed{4m - n - 2 = (2m - n - 2) + 2m}, \quad (7.8)$$

where the first summand consists of genuine alias coefficients synthesizing the zero function and the second consists of nonzero local up-splines invisible at the $n + 1$ interpolation nodes.

Proof. If $c \in \operatorname{Ker} B_n$, then $\mathcal{E}_n \mathcal{R}_n c = 0$, so $\mathcal{R}_n c \in \mathcal{Z}_n$. Every $z \in \mathcal{Z}_n$ has a coefficient preimage because $z \in \mathcal{V}_n$, and every such preimage belongs to $\operatorname{Ker} B_n$. The kernel of the restricted synthesis map is exactly $\operatorname{Ker} \mathcal{R}_n$. The dimension formula follows from (7.5) and [theorem 7.1](#). $\square$

7.3 Explicit 2-adic zero modes and Appell one modes

Let $\omega = e^{2\pi i/m}$. The imported dyadic alias theorem supplies the coefficient sequences

$$a_k^{(j,s)} = k^s \omega^{jk}, \quad 1 \leq j < m, \quad 0 \leq s \leq \nu_2(j). \quad (7.9)$$

Restricted to the active consecutive index set K_n , they are linearly independent and synthesize zero on $[-1, 1]$. Therefore each is a zero eigenvector of $\mathcal{P}_n$. Their number is

$$\sum_{j=1}^{m-1} (1 + \nu_2(j)) = 2m - n - 2, \quad (7.10)$$

which accounts for the first term in (7.8).

The one eigenspace has an equally explicit polynomial basis. For $0 \leq r \leq n$, define

$$s_k^{(r)} = h \mathcal{A}_r(kh). \quad (7.11)$$

Then $s^{(r)} = \mathcal{S}_n(x^r)$, so $\mathcal{P}_n s^{(r)} = s^{(r)}$. These $n + 1$ vectors are linearly independent because $\mathcal{S}_n$ is injective. We have therefore obtained a partial closed diagonalization:

sector	eigenvalue	multiplicity
sampled inverse-moment Appell polynomials	1	$n + 1$
confluent dyadic Fourier/Thue–Morse aliases	0	$2m - n - 2$
nonzero local up-splines vanishing at all nodes	0	$2m$

The last sector can be generated concretely by $c - \mathcal{P}_n c$ from any spanning family of coefficient vectors; finding a canonical closed basis adapted simultaneously to the nodes and the annihilator remains open.

7.4 What compression removes

In the $2m + n + 1$ -atom basis of [theorem 3.4](#), synthesis is injective, so the alias sector disappears. The corresponding coefficient projector still has rank $n + 1$, but its zero multiplicity is now exactly $2m$, the dimension of $\mathcal{Z}_n$. Thus alias compression changes the spectral ledger from

$$(n + 1) \oplus (2m - n - 2) \oplus (2m) \quad (7.12)$$

to

$$(n + 1) \oplus (2m). \quad (7.13)$$

It removes coefficient redundancy without changing the underlying sample-invisible local shapes.

8 Nested projector flags and dyadic detail multiplicities

The previous projector lives at one interpolation degree. Nested node sets produce an exact algebra of projectors and details.

Lemma 8.1 (Commuting nested interpolation projectors). *For nested nodes (5.5),*

$$\mathcal{I}^{(q)}\mathcal{I}^{(r)} = \mathcal{I}^{(r)}\mathcal{I}^{(q)} = \mathcal{I}^{(\min(q,r))}. \quad (8.1)$$

Proof. Assume $q \leq r$. The polynomial $\mathcal{I}^{(r)}f$ agrees with f at every node of $X^{(q)}$, so interpolating it on $X^{(q)}$ gives $\mathcal{I}^{(q)}f$. Conversely, $\mathcal{I}^{(q)}f \in \mathcal{P}_{n_q} \subset \mathcal{P}_{n_r}$, and the degree- n_r projector reproduces it. $\square$

Fix a terminal level s . Use the fixed-radius coefficient and synthesis maps $\mathcal{S}_s, \mathcal{R}_s$ constructed for the terminal degree n_s , so $\mathcal{R}_s\mathcal{S}_s = I$ on all of $\mathcal{P}_{n_s}$. Lift every lower interpolation projector to the same terminal coefficient space by

$$\hat{\mathcal{P}}_q^{(s)} := \mathcal{S}_s\mathcal{I}^{(q)}\mathcal{R}_s, \quad 0 \leq q \leq s. \quad (8.2)$$

For $q = s$, this is precisely the terminal coefficient projector of [theorem 6.3](#).

Theorem 8.2 (Coefficient-space interpolation flag). *The lifted operators satisfy*

$$\hat{\mathcal{P}}_q^{(s)}\hat{\mathcal{P}}_r^{(s)} = \hat{\mathcal{P}}_{\min(q,r)}^{(s)}, \quad \text{rank } \hat{\mathcal{P}}_q^{(s)} = n_q + 1. \quad (8.3)$$

In particular, every $\hat{\mathcal{P}}_q^{(s)}$ is idempotent. Define the detail operators

$$\hat{\Delta}_0^{(s)} = \hat{\mathcal{P}}_0^{(s)}, \quad \hat{\Delta}_q^{(s)} = \hat{\mathcal{P}}_q^{(s)} - \hat{\mathcal{P}}_{q-1}^{(s)} \quad (q \geq 1). \quad (8.4)$$

Then the $\hat{\Delta}_q^{(s)}$ are pairwise annihilating idempotents and

$$\text{rank } \hat{\Delta}_0^{(s)} = n_0 + 1, \quad \text{rank } \hat{\Delta}_q^{(s)} = n_q - n_{q-1} \quad (q \geq 1). \quad (8.5)$$

Proof. Because $\mathcal{I}^{(r)}\mathcal{R}_s$ takes values in $\mathcal{P}_{n_s}$,

$$\hat{\mathcal{P}}_q^{(s)}\hat{\mathcal{P}}_r^{(s)} = \mathcal{S}_s\mathcal{I}^{(q)}(\mathcal{R}_s\mathcal{S}_s)\mathcal{I}^{(r)}\mathcal{R}_s = \mathcal{S}_s\mathcal{I}^{(q)}\mathcal{I}^{(r)}\mathcal{R}_s,$$

and [lemma 8.1](#) gives the product rule. Surjectivity onto $\mathcal{S}_s(\mathcal{P}_{n_q})$ follows by applying the operator to $\mathcal{S}_sp$ with $p \in \mathcal{P}_{n_q}$, proving the rank. The detail identities follow by expanding products with the nested projector law. Their ranks are the differences of the nested ranges. $\square$

Corollary 8.3 (Dyadic spectral multiplicities). *For the nested Chebyshev–Lobatto or equispaced dyadic degree sequence $n_q = 2^q$, every detail idempotent with $q \geq 1$ has eigenvalue 1 of multiplicity*

$$n_q - n_{q-1} = 2^{q-1} \quad (8.6)$$

and eigenvalue 0 on the remaining terminal coefficient dimensions. Thus the closed loop carries an exact dyadic spectral staircase that mirrors the doubling of the interpolation spaces.

At the function level, $\mathcal{I}^{(q)} - \mathcal{I}^{(q-1)}$ is the polynomial detail in (5.7); at the coefficient level, $\hat{\Delta}_q^{(s)}$ isolates the same detail inside one terminal up-dictionary. This is the natural transform viewpoint suggested by the problem statement.

9 Convergence, Runge behavior, and coordinate stability

9.1 An exact transfer theorem

The most important analytic consequence of the loop is also the simplest: the shift expansion does not alter the polynomial interpolant.

Theorem 9.1 (Approximation behavior is inherited exactly). *Let*

$$U_n(x) := \sum_{k \in K_n} c_{n,k} u(x - kh) \quad (9.1)$$

with the coefficients in (5.4). Then

$$U_n = \mathcal{I}_n u \quad \text{pointwise on } [-1, 1]. \quad (9.2)$$

Consequently, for every norm or seminorm on functions on $[-1, 1]$,

$$\|u - U_n\| = \|u - \mathcal{I}_n u\|. \quad (9.3)$$

For a nested series, the partial sum through level q equals $\mathcal{I}^{(q)} u$. Thus convergence, divergence, endpoint oscillation, and Runge amplification are exactly those of the original interpolation sequence.

Proof. This is [theorem 5.1](#) and the telescoping identity (5.6). □

Corollary 9.2 (There is no hidden truncation error at a fixed level). *At degree n :*

1. *the inverse-moment differential series terminates after derivative order n ;*
2. *the global translate sum truncates to K_n exactly by compact support;*
3. *the substitution of cardinals into the interpolant is finite.*

Hence the only mathematical approximation error is $u - \mathcal{I}_n u$. Floating-point cancellation is a separate numerical issue.

This rules out a tempting but incorrect interpretation: replacing each Lagrange cardinal by up-atoms does not regularize a bad interpolation grid. It merely changes coordinates in the same finite-dimensional polynomial space.

9.2 Chebyshev–Lobatto convergence

Let $E_n(u) = \inf_{p \in \mathcal{P}_n} \|u - p\|_\infty$ and let Λ_n denote the Lebesgue constant of the nodes. The standard estimate is

$$\|u - \mathcal{I}_n u\|_\infty \leq (1 + \Lambda_n) E_n(u). \quad (9.4)$$

For Chebyshev–Lobatto nodes, $\Lambda_n = (2/\pi) \log n + \mathcal{O}(1)$. Since $u \in C^\infty[-1, 1]$, Jackson-type estimates imply that for every r ,

$$E_n(u) = \mathcal{O}_r(n^{-r}). \quad (9.5)$$

Corollary 9.3 (Superalgebraic stable-node loop). *For the Chebyshev–Lobatto loop and every $s > 0$,*

$$\|u - U_n\|_\infty = \mathcal{O}_s(n^{-s}). \quad (9.6)$$

The nested degrees $n_q = 2^q$ therefore produce a convergent self-shift series (5.9) with superalgebraically decreasing function error.

Proof. Choose $r > s$ in (9.5), combine it with (9.4), and use [theorem 9.1](#). □

The convergence need not be geometric. The Fabius/up function is nowhere analytic in the relevant sense [1], and the repository’s Fourier-decay volumes instead exhibit a log-Gaussian, periodically modulated decay structure. Determining the sharp Chebyshev coefficient asymptotics is an attractive bridge between those Fourier results and the present loop.

9.3 Equispaced Runge behavior

For equispaced nodes, the Lebesgue constant is exponentially large; in its classical asymptotic form,

$$\Lambda_n \sim \frac{2^{n+1}}{en \log n}. \quad (9.7)$$

The dyadic-comb volume documents severe endpoint growth for the Fabius and half-comb up interpolants, together with exact rational data and barycentric checks. The computations in [section 10](#) show the same effect for the symmetric full-interval up interpolant: the sampled uniform error is about 4.50×10^2 at degree 32 and 1.05×10^{10} at degree 64. By [theorem 9.1](#), the corresponding up-shift loop has exactly those errors. The atoms remain smooth and compactly supported; it is the polynomial projector, not the atom, that creates the boundary oscillation.

This conclusion is compatible with the general obstruction to fast stable recovery from equispaced samples [\[17\]](#). It does not assert that every equispaced method must fail; rather, it identifies the present loop exactly with the unstable global polynomial interpolant.

9.4 Lebesgue growth must appear in the decoder

Let $V(D_n)$ be the maximum row ℓ^1 norm from [\(6.20\)](#). The synthesis map obeys

$$\|\mathcal{R}_n c\|_\infty \leq \|c\|_\infty \sum_{k \in K_n} u(x - kh) = m \|c\|_\infty, \quad (9.8)$$

and the actual amplitude matrix is $C_n = h D_n$. Therefore

$$\|\mathcal{I}_n\|_{\ell^\infty(X_n) \rightarrow L^\infty[-1,1]} \leq m \|C_n\|_{\infty \rightarrow \infty} = V(D_n). \quad (9.9)$$

The left-hand side is exactly Λ_n .

Proposition 9.4 (Decoder variation dominates the Lebesgue constant).

$$\boxed{V(D_n) \geq \Lambda_n.} \quad (9.10)$$

Also,

$$h \|D_n\|_{\infty \rightarrow 1} \geq \Lambda_n, \quad (9.11)$$

because $\|\mathcal{R}_n\|_{1 \rightarrow \infty} = 1$.

The inequalities show that Runge instability cannot disappear into a benign coefficient transform. For equispaced nodes, some signed decoder norm must be exponentially large. They also expose a subtler fact: even Chebyshev interpolation can have very large fixed-radius coefficients, because D_{kj} evaluates a differential transform of the cardinal polynomial at points kh extending almost to ± 2 . This is an extrapolation problem. The Chebyshev cardinal functions are stable on $[-1, 1]$ but grow exponentially outside it, with the natural exterior scale $2 + \sqrt{3}$ at $x = 2$. Thus function-space stability and coordinate-space stability are distinct.

9.5 Computational complexity

For degree n , the direct no-dilation dictionary has $4 \cdot 2^n - 1$ atoms; the alias-compressed basis has $2^{n+1} + n + 1$. The Appell differential operator itself costs only polynomially many arithmetic operations, but writing all fixed-radius amplitudes is necessarily output-exponential in the uniform-lattice construction. By contrast, the original dilated endpoint basis of the synthesis volume uses only $n + 2$ atoms on one normalized cell. The fixed-radius condition therefore trades geometric uniformity of the atom for combinatorial explosion of the shift grid.

The output-sensitive algorithm in the repository remains useful after physical rescaling: it computes the compressed coefficients without constructing the dense alias family. Still, the output contains $2^{n+1} + n + 1$ amplitudes on a length-two interval, so no algorithm can print it in polynomial time in n .

10 Numerical experiments

The accompanying program `self_representation_experiments.py` is self-contained and uses no network access. It combines exact rational arithmetic with an independent Fourier evaluator. The period-two cosine series is

$$u(x) = \frac{1}{2} + \sum_{r \geq 0} \Phi((2r+1)\pi) \cos((2r+1)\pi x), \quad \Phi(t) = \prod_{j \geq 1} \frac{\sin(t/2^j)}{t/2^j}, \quad |x| \leq 1. \quad (10.1)$$

The experiments use 256 odd modes; the absolute sum of the next 256 coefficients is approximately 1.37×10^{-16} .

10.1 Exact rational right-inverse checks

For equispaced degrees $n = 1, 2, 4, 8$, both the interpolation nodes and the spacing- 2^{-n} shifts are dyadic. The code evaluates all needed up-values as exact fractions using the repository’s terminating highest-bit recursion for $F(a/2^q)$, constructs D_n from exact Lagrange polynomials and exact γ_r , and multiplies $\tilde{B}_n D_n$ over $\mathbb{Q}$.

Table 1: Exact rational verification. Here $V(D_n)$ is the largest decoder row ℓ^1 norm, and the last column is the lower bound $1/\text{TV}$ from adjacent kernel rows.

n	$m = 2^n$	shifts	$\tilde{B}_n D_n = I$	$V(D_n)$	adjacent $1/\text{TV}$
1	2	7	exact	1.5000	1.0000
2	4	15	exact	4.9028	1.1901
4	16	63	exact	86.2516	2.0309
8	256	1023	exact	18281.3941	4.0022

At degree 8, every one of the 81 entries of the 9×9 product is a sum of 1023 exact rational products. The result is the identity matrix with no tolerance. The decoder entries have numerators of at most 98 bits and denominators of at most 89 bits in the unreduced diagnostic table.

The adjacent-row total variations are 0.8402778, 0.4924045, and 0.2498617 at degrees 2, 4, 8 respectively, approaching the spacing $2/n$ predicted by (6.23). The canonical decoder variation is vastly larger than the universal overlap bound, especially near the extreme shift indices $k = \pm(2m-1)$; this is the extrapolation component of the transform.

10.2 Projector obliquity without forming the projector

The matrix $P_n = D_n \tilde{B}_n$ is $(4 \cdot 2^n - 1)$ -square, already 16383-square at degree 12. It is never formed. Instead, write thin QR factorizations

$$D_n = Q_D R_D, \quad \tilde{B}_n^T = Q_B R_B. \quad (10.2)$$

The nonzero singular values of P_n are exactly those of the small $(n+1)$ -square matrix $R_D R_B^T$.

The smallest nonzero singular value is numerically close to one in all these runs, while the largest singular value grows. This is typical for oblique projectors: the bad geometry is concentrated in a small number of nearly parallel range/kernel directions, not in an eigenvalue instability.

10.3 The loop inherits interpolation error

The interpolation experiment uses exact dyadic sample values and 140-digit barycentric arithmetic for the equispaced degrees 32 and 64. Chebyshev values are evaluated by the stable barycentric formula [15]. The maxima are sampled and refined, not certified continuum extrema.

Table 2: Euclidean norm of the exact coefficient projector. The displayed right-inverse residuals in the generated CSV are between 10^{-16} and 3×10^{-10} over this range.

n	shifts	$\ P_n\ _2$ equispaced	$\ P_n\ _2$ Chebyshev	ratio
2	15	2.1876	2.1876	1.00
4	63	6.6567	5.0983	1.31
6	255	33.0295	11.0067	3.00
8	1023	131.8838	40.4886	3.26
10	4095	529.9628	66.2396	8.00
12	16383	3017.1056	111.3656	27.09

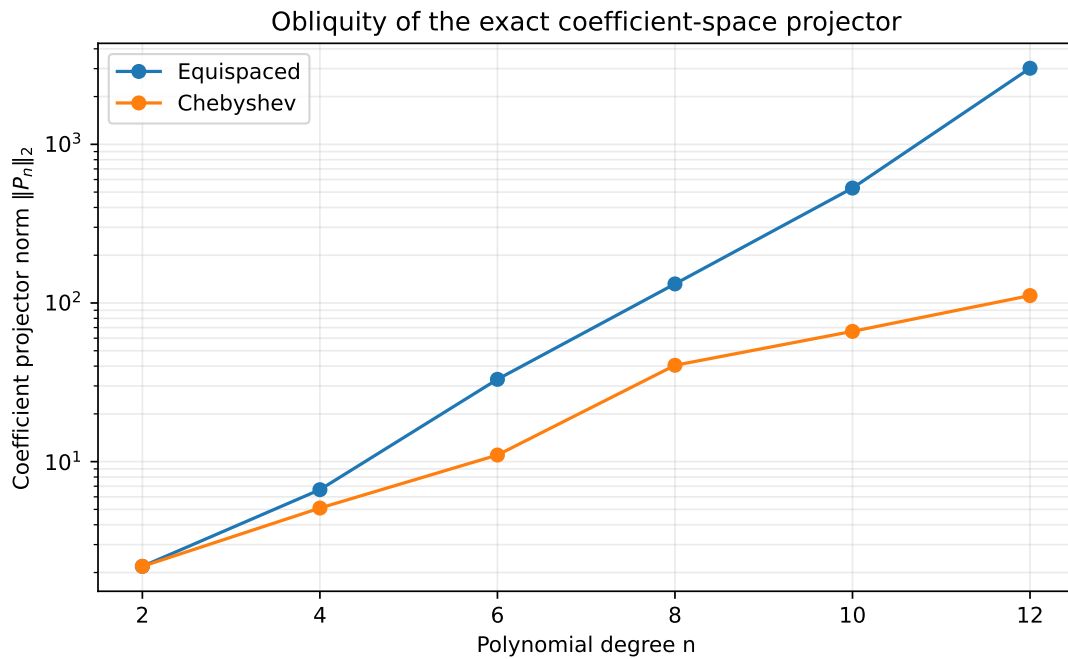


Figure 1: The projector has only the algebraic eigenvalues 0 and 1, yet its largest singular value grows rapidly. Equispaced nodes make the range and kernel almost tangent much faster than Chebyshev–Lobatto nodes.

Table 3: Sampled uniform error for interpolation of u on $[-1, 1]$. By [theorem 9.1](#), these are also exactly the errors of the corresponding up-shift loop.

n	equispaced error	Chebyshev–Lobatto error
4	7.1972×10^{-2}	8.5439×10^{-2}
8	2.0217×10^{-1}	3.5754×10^{-2}
16	1.9693×10^{-1}	2.5512×10^{-3}
32	4.4974×10^2	4.3714×10^{-4}
64	1.0463×10^{10}	4.9082×10^{-6}
128	—	2.6002×10^{-7}

The high-precision equispaced peaks occur at approximately $|x| = 0.9859694$ for $n = 32$ and $|x| = 0.9938615$ for $n = 64$, consistent with a boundary layer of width $\mathcal{O}(1/n)$. The Chebyshev errors are not perfectly monotone at low degree because u has intricate nonanalytic fine structure, but the long-range decay is clear.

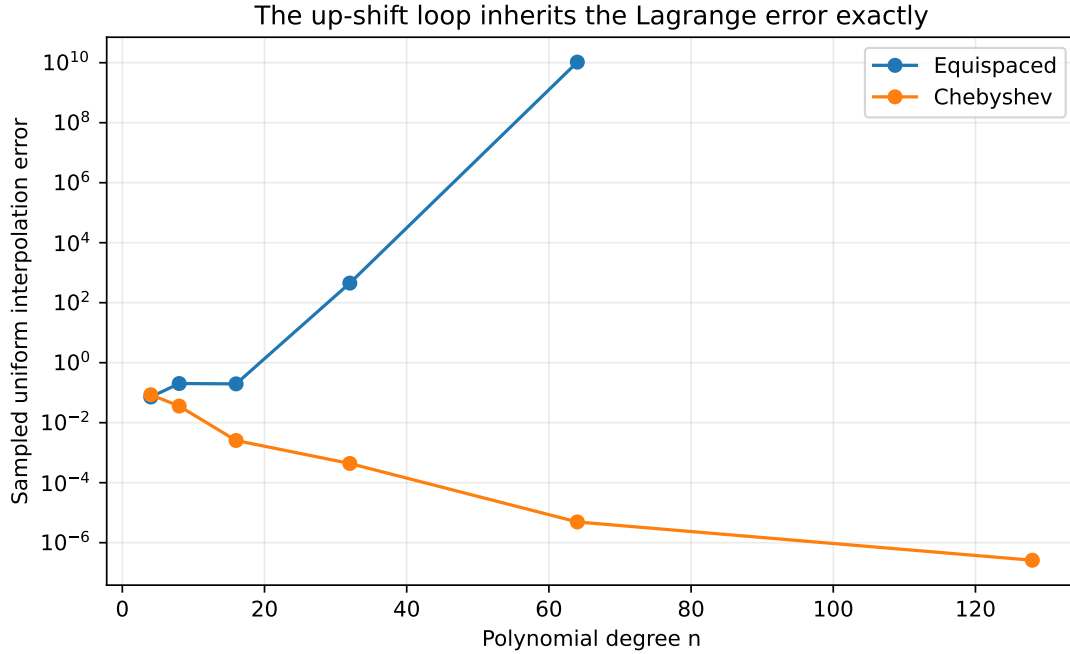


Figure 2: Function-space behavior of the closed loop. The shift synthesis neither improves nor worsens these curves mathematically: every point is the error of the original interpolation projector.

10.4 Reproducibility files

Running

```
python self_representation_experiments.py
```

regenerates:

- `results/exact_right_inverse_checks.csv`;
- `results/projector_obliquity.csv`;
- `results/interpolation_errors.csv`;
- `results/metadata.json`;
- the two PDF and PNG figures in `figures/`.

The command-line switch `-quick` omits the degree-12 matrices and the high-precision degree-64 endpoint scan.

11 Further deductions, conjectures, and research directions

The exact factorization changes the shape of the remaining questions. The basic existence problem is solved; the frontier concerns optimal gauges, conditioning, asymptotics, and structure.

11.1 Minimal fixed-radius complexity

Problem 11.1 (Arbitrary-shift fixed-radius complexity). *Let $N_n^{(1)}$ be the least number of translates $u(x - \tau_r)$, with arbitrary real shifts but no dilation, whose span on $[-1, 1]$ contains $\mathcal{P}_n$. Determine the asymptotic growth of $N_n^{(1)}$.*

The general nonpolynomial-atom dimension argument in the synthesis volume gives $N_n^{(1)} \geq n + 2$. The present uniform-lattice construction and alias compression give

$$N_n^{(1)} \leq 2^{n+1} + n + 1. \quad (11.1)$$

For uniform integer-ratio lattices, [proposition 3.5](#) proves that the exponential spacing is unavoidable. What is unknown is whether highly nonuniform shifts can collapse the count toward the linear lower bound.

Conjecture 11.2 (No polynomial-size fixed-radius universal dictionary). *There exists $c > 1$ such that $N_n^{(1)} \geq c^n$ for all sufficiently large n .*

This conjecture is intentionally labeled speculative. A possible route is to combine local non-elementarity with zero-count or Fourier-exponential-type bounds for finite sums of translates. A counterexample with $O(n)$ carefully placed centers would be equally interesting and would reveal a nonuniform analogue of the Rvachev endpoint basis.

11.2 Asymptotics of projector obliquity

The equispaced numerical values in [table 2](#) are compatible with $\log_2 \|\mathbf{P}_n\|_2 = n + o(n)$, whereas the Chebyshev sequence is much smaller over the observed range.

Conjecture 11.3 (Equispaced projector exponent). *For equispaced nodes,*

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log_2 \|\mathbf{P}_n\|_2 = 1. \quad (11.2)$$

For nested Chebyshev–Lobatto nodes, $\log \|\mathbf{P}_n\|_2 = o(n)$.

A proof should separate three mechanisms: the classical Lebesgue growth, the Appell deconvolution symbol $1/M_u$, and exterior evaluation at $kh \approx \pm 2$. The singular-value problem is only $(n + 1)$ -dimensional after reduced QR, so degrees far beyond those in this report are computationally accessible with matrix-free summation and asymptotic quadrature.

11.3 Alias gauges and minimum-variation exact synthesis

The canonical coefficient vector $\mathcal{S}_n p$ is only one representative of the affine family

$$\{c : \mathcal{R}_n c = p\} = \mathcal{S}_n p + \text{Ker } \mathcal{R}_n. \quad (11.3)$$

The explicit confluent Fourier vectors [\(7.9\)](#) make this a structured gauge freedom.

Problem 11.4 (Optimal exact gauge). *For a chosen norm or weighted norm, find*

$$c_p^* = \arg \min \{\|c\| : \mathcal{R}_n c = p\} \quad (11.4)$$

and express the minimizing correction to $\mathcal{S}_n p$ through the q -integer annihilator $A_n(E)$ or its confluent Fourier basis. Determine whether a symmetry-preserving gauge can reduce the exponential coefficient growth while retaining exact continuum equality.

This must not be confused with the Moore–Penrose right inverse of the nodal matrix B_n . The latter gives an orthogonal coefficient projector of norm one but guarantees only nodal agreement; it need not synthesize the interpolation polynomial between nodes. The meaningful optimization is inside [\(11.3\)](#).

The sign theorem and total-variation bound remain constraints on any row-unital decoder. One can formulate a convex program minimizing maximum negative mass, maximum row variation, or an ℓ^2 energy while imposing both exact polynomial reproduction and symmetry.

11.4 A continuum limit of the signed Markov decoder

The pair $(\tilde{B}_n, D_n)$ resembles a discrete smoothing kernel and its finite-order inverse. However, the grid $h = 2^{-n}$ becomes exponentially finer than the node spacing, so the matrices are strongly rectangular.

Research question 11.5 (Renormalized decoder limit). *Does a rescaling of the row index near the central region or near $kh = \pm 2$ produce a weak limit of the signed measures*

$$\mu_{n,k} = \sum_{j=0}^n D_{kj}^{(n)} \delta_{x_j} ? \quad (11.5)$$

Can its central limit be described by the inverse differential symbol $1/M_u$, while its edge limit is controlled by polynomial extrapolation?

For equispaced nodes, the universal lower bound $V(D_n) \gtrsim n/2$ comes from the total variation of neighboring translated up densities. The much larger observed variation suggests a two-scale decomposition: a local deblurring cost and an exterior cardinal-growth cost.

11.5 q -binomial minors and Thue–Morse spectral algebra

The determinant identity (6.18) and the zero-mode basis (7.9) invite a more explicit q -combinatorial treatment.

Problem 11.6 (Minor expansion through the annihilator). *Express the minors of D_n after quotienting by $\text{Ker } \mathcal{R}_n$ in terms of coefficients of*

$$A_n(z) = \prod_{r=1}^n [2^r]_z \quad (11.6)$$

and, where possible, Gaussian binomial coefficients or restricted binary partition numbers. Determine whether the Cauchy–Binet sum admits a sign-reversing involution leaving a small positive residue.

The exact multiplicity $2m - n - 2 = \sum_{j=1}^{m-1} (1 + \nu_2(j))$ already identifies the projector’s alias zero spectrum with the dyadic zero divisor of $\hat{u}$. A closed basis for the remaining $2m$ nodal-zero modes might combine Lagrange vanishing factors with the same cyclotomic blocks.

11.6 Lambert- W and the Runge boundary layer

The endpoint values of F and u are governed in the repository by Lambert- W saddle inversions and log-periodic corrections. Equispaced interpolation introduces a second endpoint saddle through binomial barycentric weights. The observed peak positions satisfy $n(1 - |x_{\max}|) \approx 0.45$ at $n = 32$ and 0.39 at $n = 64$.

Conjecture 11.7 (Boundary-layer scaling with log-periodic correction). *For dyadic equispaced degrees, the dominant symmetric-up Runge peak lies at*

$$1 - |x_{\max,n}| = \frac{\alpha(\log_2 n) + o(1)}{n}, \quad (11.7)$$

where α is bounded and may carry a small periodic modulation inherited from the dyadic endpoint asymptotics. The logarithm of the peak amplitude has an expansion obtained by coupling the binomial saddle with the Lambert- W_{-1} endpoint saddle of u .

A viable derivation would insert the all-orders small-argument expansion of F into the Newton forward form of the equispaced interpolant, then perform a two-parameter saddle analysis in the node index and boundary-layer coordinate. This directly connects the inverse-Fabius/Lambert- W corpus to the present transform.

11.7 Node sets co-designed with the up kernel

Classical node selection minimizes polynomial interpolation growth. The factorization suggests a stronger objective involving both the Lagrange space and the up dictionary, for example

$$\min_{X_n} \left\| D_n(X_n) \tilde{B}_n(X_n) \right\|_2, \quad \min_{X_n} V(D_n(X_n)), \quad \text{or} \quad \max_{X_n} \theta_n. \quad (11.8)$$

The minimizers need not be exactly Chebyshev points: the positive kernel $u(x_i - kh)$ introduces a scale-dependent geometry absent from ordinary Lebesgue optimization. Up-adapted Fekete or Leja points are a promising computational frontier.

11.8 Formalization

Most identities after [theorem 3.1](#) are finite-dimensional and well suited to Lean:

1. define the terminating operator $\mathcal{A}$ on `Polynomial`;
2. import or formalize the fixed-radius affine conjugation of the existing synthesis theorem;
3. construct the Lagrange basis and prove $B_n C_n = I$;
4. derive idempotence, rank, direct sums, and the nested projector algebra;
5. formalize the Markov-overlap sign theorem independently of analysis;
6. connect the alias kernel to the existing Thue–Morse and valuation lemmas.

The exact degree-2, 4, and 8 rational checks can serve as executable regression certificates.

12 Conclusion

The requested loop closes cleanly once the synthesis theorem is viewed through affine conjugation. Degree- n polynomial reproduction at scale 2^n is the same statement as fixed-radius reproduction by spacing- 2^{-n} translates. Applying the resulting inverse-moment Appell sampler to every Lagrange cardinal yields an explicit finite matrix D_n ; applying it after interpolation yields the canonical self-amplitudes

$$c_{n,k} = 2^{-n} [\mathbf{M}_u(D)^{-1} \mathcal{I}_n u](k2^{-n}).$$

The apparent self-representation operator is exactly the ordinary Lagrange projector in different coordinates.

That exactness is both the strength and the warning. It provides a right inverse, a Markov/signed-Markov factorization, a complete two-point spectrum, a 2-adic alias multiplicity formula, an exact local direct sum, and a nested dyadic spectral flag. It also proves that the transformation cannot suppress Runge’s phenomenon: stable nodes remain stable, unstable nodes remain unstable, and every fixed-level truncation is already exact. The new mathematics lies in the coefficient geometry—alias gauges, oblique angles, negative mass, extrapolation growth, and the distinction between nodal and continuum exactness.

The most promising next step is to optimize within the explicit alias nullspace. A minimum-variation, symmetry-preserving exact gauge could retain the conceptual elegance of literal up shifts while substantially reducing the enormous canonical coefficients. In parallel, the equispaced boundary layer offers a concrete meeting point for the repository’s Lagrange, Thue–Morse, Fourier-decay, and Lambert- W asymptotic machinery.

A Formula sheet

For quick reference, the complete canonical transform is:

$$m = 2^n, \quad h = 2^{-n}, \quad (\text{A.1})$$

$$K_n = \{-2m + 1, \dots, 2m - 1\}, \quad (\text{A.2})$$

$$\frac{1}{M_u(z)} = \exp \left[- \sum_{r \geq 1} \frac{B_{2r}}{2r(1 - 4^{-r})} \frac{z^{2r}}{(2r)!} \right] = \sum_{r \geq 0} \gamma_r \frac{z^r}{r!}, \quad (\text{A.3})$$

$$\mathcal{A}p = \sum_{r=0}^{\lfloor \deg p/2 \rfloor} \frac{\gamma_{2r}}{(2r)!} p^{(2r)}, \quad (\text{A.4})$$

$$D_{kj}^{(n)} = (\mathcal{A}\ell_j)(kh) = \frac{1}{\omega'_n(x_j)} \sum_{r=0}^{\lfloor n/2 \rfloor} \gamma_{2r} e_{n-2r}(\{kh - x_s : s \neq j\}), \quad (\text{A.5})$$

$$\ell_j(x) = h \sum_{k \in K_n} D_{kj}^{(n)} u(x - kh), \quad (\text{A.6})$$

$$c_k(y) = h \sum_{j=0}^n D_{kj}^{(n)} y_j, \quad (\text{A.7})$$

$$\mathcal{L}_n y = \sum_{k \in K_n} c_k(y) u(\cdot - kh), \quad (\text{A.8})$$

$$\tilde{B}_{ik} = hu(x_i - kh), \quad \tilde{B}_n D_n = I_{n+1}, \quad P_n = D_n \tilde{B}_n, \quad (\text{A.9})$$

$$\chi_{P_n}(\lambda) = \lambda^{4 \cdot 2^n - n - 2} (\lambda - 1)^{n+1}. \quad (\text{A.10})$$

B Dimension and multiplicity ledger

Table 4: Exact dimensions for degree n , $m = 2^n$.

Object	Dimension	Interpretation
Dense fixed-radius coefficient space	$4m - 1$	all active $h\mathbb{Z}$ shifts on $[-1, 1]$
Local up-shift function space $\mathcal{V}_n$	$2m + n + 1$	imported multi-cell dimension
Polynomial subspace $\mathcal{P}_n$	$n + 1$	one-eigenspace of the loop projector
Nodal-zero local space $\mathcal{Z}_n$	$2m$	nonzero up-splines invisible at nodes
Dense alias kernel $\text{Ker } \mathcal{R}_n$	$2m - n - 2$	confluent dyadic Fourier modes
Dense sample kernel $\text{Ker } B_n$	$4m - n - 2$	full zero-eigenspace of P_n
Compressed coefficient basis	$2m + n + 1$	alias-free right-end shift basis
Compressed projector zero eigenspace	$2m$	only the nodal-zero sector remains

C Notes on numerical verification

The exact arithmetic path is deliberately independent of the Fourier path. For $n = 1, 2, 4, 8$:

1. γ_r are generated symbolically from the Bernoulli cumulant series;

2. Lagrange cardinals are multiplied over `Fraction`;
3. D_n is evaluated by exact derivatives and Horner evaluation;
4. Fabius values at dyadic rationals are generated by the terminating highest-bit recursion;
5. $u(t) = F(1 - |t|)$ evaluates every entry of $\tilde{B}_n$ exactly;
6. the matrix product is performed over $\mathbb{Q}$.

The Fourier cosine evaluator is used only for non-dyadic floating-point nodes, plots, and larger projector diagnostics. This separation makes it unlikely that a shared implementation error could manufacture $\tilde{B}_n D_n = I$.

D Selected corpus crosswalk

Table 5: Repository material used to set notation and avoid duplication.

Corpus branch	Role in this report
Primary exposition	Definitions of F and u , functional equations, probability model, sinc product, moments, dyadic arithmetic.
Up polynomial synthesis	Imported Strang–Fix scale classification, global Appell deconvolution, local dimension, alias annihilator, endpoint and multi-cell bases, fast coefficient algorithm.
Dyadic-comb frontiers	Imported Lagrange/Hermite interpolation setup, symmetry reductions, Lebesgue/Runge analysis, exact dyadic numerical data.
Representation frontiers	Fourier, Faber–Schauder, Bernstein, Bessel, and multiresolution formulas; independent cosine evaluator used by the experiments.
Thue–Morse atlas/frontiers	Prouhet cancellation, dyadic products, finite block calculus, valuation multiplicities.
Exponent and q -series frontiers	q -Pochhammer, Gaussian-binomial, Bernoulli–Bell expansions, finite sinc approximants and transport geometry.
Inverse and sampling frontiers	Inverse-Fabius asymptotics, sampling operators, endpoint structure, Lambert- W inversion.
Spectra and arithmetic frontiers	Spectral multiplicities, exact rational structures, and matrix viewpoints compared against the projector results.
Fourier-decay corpus	Log-Gaussian envelope and periodic modulation relevant to sharp Chebyshev and boundary-layer asymptotics.
Lean walkthrough	Existing formal definitions and theorem dependency graph for future formalization.
Paper transcriptions	Historical and published sources on Fabius/up arithmetic, lacunary products, and related functional equations.
Archive	Frozen derivations and predecessor reports used for provenance and to avoid re-claiming absorbed formulas.

References

- [1] J. Fabius, A probabilistic example of a nowhere analytic C^∞ -function, *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete* **5** (1966), 173–174.
- [2] V. L. Rvachev and V. A. Rvachev, A certain finite function, *Dopovidi Akademii Nauk Ukrains’koi RSR, Series A* (1971), 705–707, 764.

- [3] V. L. Rvachev and V. A. Rvachev, On the representation of polynomials by functions with compact support, *Mathematical Physics*, No. 11, Naukova Dumka, Kiev, 1972, 126–129.
- [4] V. A. Rvachev, On approximation by means of the function $up(x)$, *Doklady Akademii Nauk SSSR* **233** (1977), no. 2, 295–296.
- [5] V. A. Rvachev, Compactly supported solutions of functional-differential equations and their applications, *Russian Mathematical Surveys* **45** (1990), no. 1, 87–120.
- [6] J. Arias de Reyna, Definición y estudio de una función indefinidamente diferenciable de soporte compacto, *Revista de la Real Academia de Ciencias Exactas, Físicas y Naturales de Madrid* **76** (1982), 21–38; English translation arXiv:1702.05442.
- [7] J. Arias de Reyna, Arithmetic of the Fabius function, *INTEGERS* **18** (2018), Article A51.
- [8] G. Strang and G. Fix, A Fourier analysis of the finite element variational method, in *Constructive Aspects of Functional Analysis*, C.I.M.E., 1973, 793–840.
- [9] C. de Boor and A. Ron, Fourier analysis of the approximation power of principal shift-invariant spaces, *Constructive Approximation* **8** (1992), 427–462.
- [10] C. de Boor, R. A. DeVore, and A. Ron, Approximation from shift-invariant subspaces of $L_2(\mathbb{R}^d)$, *Transactions of the American Mathematical Society* **341** (1994), 787–806.
- [11] R.-Q. Jia, Shift-invariant spaces and linear operator equations, *Israel Journal of Mathematics* **103** (1998), 259–288.
- [12] S. Roman, *The Umbral Calculus*, Academic Press, 1984.
- [13] L. Comtet, *Advanced Combinatorics*, Reidel, 1974.
- [14] J.-P. Allouche and J. Shallit, *Automatic Sequences: Theory, Applications, Generalizations*, Cambridge University Press, 2003.
- [15] J.-P. Berrut and L. N. Trefethen, Barycentric Lagrange interpolation, *SIAM Review* **46** (2004), no. 3, 501–517, DOI 10.1137/S0036144502417715.
- [16] N. J. Higham, The numerical stability of barycentric Lagrange interpolation, *IMA Journal of Numerical Analysis* **24** (2004), 547–556.
- [17] R. B. Platte, L. N. Trefethen, and A. B. J. Kuijlaars, Impossibility of fast stable approximation of analytic functions from equispaced samples, *SIAM Review* **53** (2011), no. 2, 308–318, DOI 10.1137/090774707.
- [18] L. N. Trefethen, *Approximation Theory and Approximation Practice*, SIAM, 2013.
- [19] V. Reshetnikov, *The Fabius Function and Rvachev’s Up-Function*, primary exposition in the ProveIt repository, Analysis/FabiusFunction/docs, snapshot commit 4544e9af6b94820402c69967be1dff3f85e43c1e, 2026.
- [20] V. Reshetnikov, *Exact Polynomial Synthesis by Finite Rvachev Up-Function Dictionaries*, semi-formalized-research-frontiers/drafts/representations/Up_Polynomial_Synthesis/, ProveIt snapshot commit 4544e9af6b94820402c69967be1dff3f85e43c1e, 2026.
- [21] V. Reshetnikov, *Dyadic-Comb Frontiers for the Fabius–Rvachev System*, semi-formalized-research-frontiers/drafts/inverse-and-sampling/Dyadic_Comb_Frontiers/, ProveIt snapshot commit 4544e9af6b94820402c69967be1dff3f85e43c1e, 2026.

- [22] V. Reshetnikov, *Representation Frontiers for the Fabius–Rvachev System*, consolidated frontier volume in ProveIt, snapshot commit 4544e9af6b94820402c69967be1dff3f85e43c1e, 2026.
- [23] V. Reshetnikov, *Inverse and Sampling Frontiers for the Fabius–Rvachev System*, consolidated frontier volume in ProveIt, snapshot commit 4544e9af6b94820402c69967be1dff3f85e43c1e, 2026.